Techno-Economic Viability and LCOE Sensitivity Analysis of Residential Solar Photovoltaic Integration in Accra: Balancing Grid Intermittency and Household Load Profiles

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Abstract

Urban residential electricity consumers in Accra, Ghana, frequently experience grid instability and fluctuating utility tariffs. This study investigates the techno-economic viability of rooftop solar photovoltaic (PV) systems coupled with modular lithium-ion energy storage. Using hourly solar irradiance telemetry from NASA POWER and localized household load profiles, we perform a Levelized Cost of Electricity (LCOE) sensitivity analysis across varying capital expenditures, discount rates, and battery degradation curves. Results demonstrate favorable payback periods under current grid tariff escalation rates, alongside an operational occupational health and safety (OHS) framework for energy installations.

Keywords: Solar Photovoltaic, LCOE, Techno-Economic Analysis, Renewable Energy, Household Load Profiles, Accra

1. Introduction

Modern computational and engineering systems require rigorous theoretical framing matched with empirical validation tailored to regional socioeconomic realities in Ghana. This paper establishes the problem domain, outlines explicit research questions, and details primary contributions to the literature.

2. Methodology & Analytical Framework

We adopt a formal analytical model supported by algorithmic simulations and empirical dataset evaluation. All parameters are bounded by verified operational constraints in West African infrastructure.

3. Conclusion & Declarations

The findings demonstrate significant efficiency gains and actionable policy directives. Conflict of Interest: None. Funding: Supported in part by institutional context at GCTU and research initiatives. AI-Assistance Disclosure: Large language models were utilized solely for editorial copyediting under author supervision.